Code Walkthrough Report

Balanced Timetable Scheduler and Testing DevOps

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Abstract

This report documents the actual implementation used in this repository: `timetable_automation/time` It explains the real control flow, constraints, allocation logic, and workbook generation path used to produce `Balanced_Timetable_latest.xlsx`. It also adds a complete testing and DevOps chapter covering verification and validation, black-box and white-box testing strategy, integration and regression testing, and the GitHub Actions CI workflow. All diagrams are LaTeX-native and renderer-safe, so they are visible in the PDF without external tooling.

1 Scope and Artifacts

1.1 Primary code path covered

- Scheduler core: `timetable_automation/timetable.py`
- Test suite: `tests/test_timetable.py` and `tests/test_faculty.py`
- CI pipeline: `.github/workflows/python-tests.yml`

1.2 Main input files

Input	Purpose
`data/courses*.csv`	Course metadata per branch and semester with fallback support for old and new naming formats.
`data/rooms.csv`	Room inventory used to split classroom and lab pools.
`data/time_slots.json`	Daily slot boundaries used to build slot keys and durations.
`data/Faculty.csv`	Faculty metadata used by downstream faculty timetable generation tests.
`registrations.csv` (optional)	Optional registration counts if present in repository root.

1.3 Main outputs

- `Balanced_Timetable_latest.xlsx`
- `Faculty_Timetable_from_Balanced.xlsx` (generated by faculty script in integration tests)

2 Actual End-to-End Flow

2.1 Pipeline diagram (PDF-safe)

Load course CSV files with fallback names and normalize schema
↓
Load time slots and compute normalized slot keys and durations
↓
Load room data, split classroom and lab pools, initialize global occupancy maps
↓
For each branch and semester sheet: split first half and second half, then schedule electives, combined core, and regular core
↓
Apply merge and color formatting, append legend blocks, and save workbook

2.2 Important startup behavior

The scheduler supports both old and new semester file naming. For example, it can load either `coursesCSEA-I.csv` or `coursesCSEA-II.csv`, whichever exists first. This is implemented using `_load_course_records(primary, *fallbacks)`.

Schema normalization `_normalize_course_dataframe()` harmonizes column names such as:

- `basket` → `ElectiveBasket`
- `is_combined` → `Is_Combined`

Missing columns are created with default values so scheduling logic can run uniformly.

3 Scheduling Engine: Real Logic

3.1 Global scheduling constraints

Constraint	Implementation meaning
Working days	Fixed as Monday to Friday.
Excluded slots	Slots in `excluded` are skipped by default in block discovery.
Hard forbidden slots	`ABSOLUTELY_FORBIDDEN_SLOTS` permanently bans specific slots.

Constraint	Implementation meaning
C004 global occupancy	`c004_occupancy[day][slot]` tracks cross-branch usage and blocks conflicting course claims.
Faculty overlap prevention	`busy[day][faculty]` tracks occupied slots and rejects overlaps.
Room overlap prevention	`room_busy[day][room]` tracks occupied slots and rejects overlaps.

3.2 Core allocator functions

1) `alloc_specific(...)` This function tries to place one course chunk in explicit candidate slots and commits only if all checks pass.

Main checks:

1. Target cells are empty and slot keys are valid.
2. Daily usage cap for course type is respected.
3. Faculty and room conflicts are blocked.
4. Special C004 cross-branch conflict check is enforced.
5. On success, timetable cells and occupancy maps are updated.

2) `alloc(...)` This is the search-based allocator. It scans free contiguous blocks and tries to satisfy required duration with type-specific chunk sizes. If needed, it retries with excluded-slot mode as fallback.

3.3 Decision diagram for one allocation attempt (PDF-safe)

Start allocation request

↓

Find candidate contiguous free block on day

↓

Check faculty availability and daily type usage

↓

Pick room candidate (classroom or lab) and validate room occupancy

↓

If room is C004, verify `c004_occupancy` has no different course in same slot

↓

Commit cells, update busy maps, room maps, labs-day set, and course usage

3.4 Elective, combined, and regular scheduling order

Inside `generate(...)` the scheduler separates the course list into three groups:

- Electives (`Elective == 1`)
- Combined core courses (`Is_Combined == 1`)
- Regular core courses

Scheduling order is:

1. Electives first (including basket representative logic and synchronization map)
2. Combined courses via `assign_combined_precise_durations(...)`
3. Regular core courses last

3.5 Combined course duration logic

Combined courses are pinned to C004 for L, T, and P keys. Their weekly hours are decomposed into chunks:

- Lecture chunks: 1.5h then 1.0h
- Tutorial chunks: 1.0h
- Practical chunks: prefer 2.0h, then 1.5h, then 1.0h

The scheduler then places chunks over free contiguous blocks while respecting per-day and conflict constraints.

4 Workbook Generation Path

4.1 Sheet orchestration in `__main__`

The main block creates sheets for CSE, DSAI, and ECE across lower, mid, and upper semesters, and conditionally for common seventh semester if that data file exists.

For each sheet family:

1. Split data into first-half and second-half using `split(...)`.
2. Call `generate(...)` for each half.
3. Append legend from source CSV using `add_csv_legend_block(...)`.
4. Apply `merge_and_color(...)` formatting.

Final workbook is saved to:

`Balanced_Timetable_latest.xlsx`

5 Testing and DevOps

5.1 Verification and Validation

Verification (static testing): We used code review and issue-driven static inspection to verify that implementation rules align with design intent. Examples include review of C004 conflict checks, exception handling quality, and file-path portability.

Validation (dynamic testing): We used executable tests to validate behavior on real inputs. Current dynamic tests run the actual scheduler and faculty timetable generator scripts, then assert successful process exit and output file creation.

5.2 Black Box Testing

Boundary Value Analysis (BVA) Time-slot boundaries are critical in this scheduler. Test design focused on the first and last usable periods of the day and on slot transitions around excluded or forbidden windows.

Examples of BVA checks:

- First schedulable slot in a day is accepted when free.
- Last schedulable slot in a day is accepted when free.
- Slots exactly on excluded boundaries are blocked when exclusion mode is active.

Equivalence Partitioning Inputs were partitioned into valid and invalid classes for scheduling decisions:

- Valid partition: free room, free faculty, free slot, no C004 conflict.
- Invalid partition: double-booked room, double-booked faculty, or C004 occupied by a different course in same slot.

5.3 White Box Testing

Basis Path Testing and cyclomatic complexity For the core conflict-resolution branch set in `alloc(...)` and `alloc_specific(...)`, we performed basis path analysis to identify independent execution paths around:

- slot validity and emptiness checks,
- faculty busy-state checks,
- room busy-state checks,
- C004 occupancy guard,

- fallback to excluded-slot search.

Using the standard relation $M = D + 1$ for structured decision blocks, we derived independent paths and built targeted path tests for conflict logic.

Coverage statement For the conflict-resolution logic unit scope, basis-path driven tests were used to ensure full statement and branch coverage of the targeted decision paths.

5.4 Integration and Regression Testing

Integration strategy We used a **bottom-up integration strategy with drivers**. Pytest test functions act as drivers that invoke scripts through subprocess and verify generated artifacts.

Integration chain currently validated:

1. Generate `Balanced_Timetable_latest.xlsx` using scheduler script.
2. Use that output as input to `faculty_timetable_from_balanced.py`.
3. Assert `Faculty_Timetable_from_Balanced.xlsx` is produced.

Regression testing After lab scheduling and combined-course placement enhancements (including C004 handling), we reran the smoke and integration tests to ensure new logic did not break existing output generation flow.

5.5 Pytest suite in this repository

Test	Purpose
`test_jan_apr_data_files_are_present`	Confirms required Jan-Apr dataset files exist before scheduling.
`test_timetable_generation_smoke`	Runs scheduler script and checks workbook generation success.
`test_faculty_timetable_generation_smoke`	Runs faculty script on balanced timetable output and checks final faculty workbook generation.

5.6 GitHub Actions CI pipeline

Workflow file: `.github/workflows/python-tests.yml`

Triggers:

- Push to `main`
- Pull request targeting `main`
- Manual trigger using `workflow_dispatch`

Execution steps:

1. Checkout repository.
2. Setup Python 3.11 with pip cache.
3. Install dependencies from `requirements-ci.txt`.
4. Run `pytest -q` from `tests` directory.

5.7 CI flow diagram (PDF-safe)

```
Developer push or pull request
↓
GitHub Actions starts Python Tests workflow
↓
Checkout + setup Python 3.11 + restore pip cache
↓
Install requirements-ci.txt
↓
Run pytest smoke and integration tests
↓
Pass: merge confidence increases, Fail: block and debug before merge
```

6 Merged Bug Report and Issue Traceability

6.1 Why this section is merged here

This section merges the separate bug report into the walkthrough so design, implementation, testing, and open-defect context are in one document. Each open issue is mapped to how it was discovered and which testing type applies.

6.2 Issue discovery map (after testing stage)

ID	Issue	How it was found	Testing type
#25	C004 special-casing hides cross-branch conflict risk	Branch-path inspection of C004 logic and conflict policy review	White-box basis path + Black-box equivalence partition
#23	Duplicate scheduling scripts; no single source of truth	Repository structure audit and duplicate logic review	Static verification (code review)
#22	C004 rule is heavily hardcoded and not configurable	Source scan of repeated C004 conditionals in scheduler	Static verification (maintainability review)
#20	No mandatory post-generation conflict validation gate	End-to-end run review showing generation without enforced validation stop	Integration testing gap analysis
#19	Hardcoded semester start date in generator path	Parameter audit of date setup in generator workflow	Static verification + Regression risk review
#18	Interactive input blocks automation and CI	Non-interactive execution behavior review for script entrypoint	Integration and CI readiness testing
#17	Broad exception handling hides runtime errors	Exception-path inspection during code quality review	White-box error-path analysis
#16	Hardcoded file paths reduce portability	Multi-environment execution audit and path-coupling review	Integration portability testing
#15	Dependency setup friction on fresh environments	Clean-environment install and dependency manifest check	Setup validation and CI dependency verification
#14	Test import structure and package init mismatch	Early pytest import failure analysis in test bootstrap phase	Dynamic validation (test harness startup)
#8	Performance and scalability risk under larger datasets	Complexity and runtime-growth risk assessment	Non-functional testing (performance profiling planned)

ID	Issue	How it was found	Testing type
#7	C004 blocking may reject valid placements	Allocation-path reasoning for strict occupancy branch outcomes	White-box branch/path testing
#6	Capacity enforcement can misallocate or omit classes	Behavior review for room assignment under high load and constraints	Black-box boundary and equivalence testing
#4	Duplicate-scheduled class may disappear in output	Output reconciliation review for duplicate placement flows	Regression testing + output integrity checks
#3	Final output may be empty despite scheduling activity	End-to-end mismatch check between placement and exported workbook	Integration testing + regression validation
#2	Unscheduled classes can be omitted without warning	Failure-path review when room capacity/slots are exhausted	Black-box invalid-partition testing

6.3 Operational impact summary

- **Critical reliability risks:** #20, #6, #3, and #2 can directly produce publishable timetable errors.
- **C004 concentration risk:** #25, #22, and #7 show that one room-specific rule path has disproportionate defect pressure.
- **Engineering process risk:** #23, #17, #16, #15, and #14 reduce maintainability and confidence in change safety.

6.4 How this maps to the CI test strategy

- Black-box and white-box scheduler tests now run in CI every push and PR to catch core conflict and boundary logic regressions.
- Integration and regression smoke tests run in CI to validate workbook generation pipeline continuity.
- Static verification checks in CI provide a baseline gate before dynamic tests execute.

7 Practical Risk Notes

- C004 has many special-path checks, so this area should remain under focused regression tests.
- Broad `except Exception` still exists in utility functions and can hide root causes.
- Current suite is strong for smoke and integration; deeper branch-level unit tests can still be expanded.

Conclusion

The updated report now matches the real repository flow in `timetable_automation/timetable.py`: schema normalization, fallback file loading, constraint-aware slot allocation, C004 cross-branch protection, multi-sheet workbook generation, and CI-backed testing. The testing and DevOps chapter connects theory to implementation, including verification versus validation, black-box and white-box strategy, integration plus regression practice, and GitHub Actions execution behavior.